

Theoretical model concept IDP

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1 Simple SIR model

We should start with SIR model based on three differential equations adjusted for modeling only the different municipalities:

$$\begin{cases} \frac{dS_i}{dt} = -\beta S_i \sum_j W_{ij} \frac{I_j}{N_j} \\ \frac{dI_i}{dt} = \beta S_i \sum_j W_{ij} \frac{I_j}{N_j} - \gamma I_i \\ \frac{dR_i}{dt} = \gamma I_i \end{cases}$$

From these equation we want to know the amount of susceptible, infected and recovered people in each municipality ($S_i, I_i, & R_i$ respectively), where we use an index for each municipality in the Netherlands (the easiest way to do this is probably by sorting them in alphabetical order). We will need the following constants and initial values to be able to start the simulation:

1.1 Constants:

- N_i list of the total amount of people in each municipality.
- d_{ij} matrix containing the distance of a municipality with each different one. (=symmetric)

1.2 Initial values:

- $S_i(t = 0) = N_i - I_i(t = 0) - R_i(t = 0)$: list with the amount of people susceptible to infection in each municipality at $t = 0$.
- $I_i(t = 0)$: list with the amount of infected people in each municipality at $t = 0$.
- $R_i(t = 0)$: list with the amount of people that immune to infection in each municipality at $t = 0$.

1.3 Simulation:

After both these and the parameters (defined later) have been set, we can start the simulation. To calculate S_i, I_i, R_i at a timestep Δt later, we first need the chance of a susceptible individual getting infected and an infected individual recovering in municipality i derived from the differential equations:

- $p_{inf,i} = 1 - e^{-\lambda_i \Delta t}$, where:
 $\lambda_i = \beta \sum_j W_{ij} \frac{I_j}{N_j}$ is the force of infection for the municipality. W_{ij} is a matrix that tracks the amount of contact a person in municipality i makes with one from j . We can define this via a distance-based model:
 $W_{ij}^{raw} = \frac{N_j^\alpha}{(d_{ij} + d_0)^\theta} \longrightarrow W_{ij} = \frac{W_{ij}^{raw}}{\sum_k W_{ik}^{raw}}$ (for normalization).
- $p_{rec,i} = 1 - e^{-\gamma \Delta t}$.

With this we can approximate the total amount of new infections in municipality i by taking a binomial draw of $(S_i, p_{inf,i}) = newinfections_i$. We can approximate the amount of new recoveries the same way, by taking a binomial draw of $(I_i, p_{rec,i}) = newrecoveries_i$. With this we can update our model:

$$\begin{cases} S_i(t + \Delta t) = S_i(t) - newinfections_i \\ I_i(t + \Delta t) = I_i(t) + newinfections_i - newrecoveries_i \\ R_i(t + \Delta t) = R_i(t) + newrecoveries_i \end{cases}$$

1.4 Parameters:

- β : transmission rate, chance of individual to get infected per contact per unit time.
- γ : recovery rate, the rate at which an infected individual recovers per unit time.
- Δt : chosen time step for our simulation.
- α : exponent that controls how much more contacts there will be in larger municipalities. $\sim [1, 1.5]$
- θ : exponent for distance decay, which determines how much a greater amount of distance between municipalities decreases the contacts between these municipalities. $\sim [1, 2]$
- d_0 : to avoid dividing by zero in the contact matrix and define how much contacts there are within a municipality itself.

1.5 Limiting assumptions:

- People mix homogeneously within each municipality and contacts between municipalities only depends on distance and population size.
- All people are the same age and ave the same chance of getting infected/recovering.
- Amount of people in each municipality is fixed, so no births, deaths or migration.
- People gain permanent immunity after recovery.
- Individuals are immediately infectious after having been infected themselves.
- Vaccinations are not present in this model.

1.6 Impact of measures:

The way measures could impact the simulation is through the parameters β and γ and through the contact matrix W_{ij} .

2 SIRVDB model

Here we have adjusted the equations to neutralize the assumptions that simplified our model too much to be able to give realistic results.

$$\begin{cases} \frac{dS_{i,a}}{dt} = -\beta S_{i,a} \sum_j \sum_b W_{ij} C_{ab} \frac{I_{j,b}}{N_{j,b}} - \nu_{i,a} S_{i,a} + \omega_{r,a} R_{i,a} + \omega_{v,a} V_{i,a} \\ \frac{dI_{i,a}}{dt} = \beta S_{i,a} \sum_j \sum_b W_{ij} C_{ab} \frac{I_{j,b}}{N_{j,b}} - (\gamma_a + \mu_a) I_{i,a} \\ \frac{dR_{i,a}}{dt} = \gamma_a I_{i,a} - \omega_{r,a} R_{i,a} \\ \frac{dD_{i,a}}{dt} = \mu_a I_{i,a} \\ \frac{dV_{i,a}}{dt} = \nu_{i,a} S_{i,a} - \omega_{v,a} V_{i,a} \end{cases}$$

A new subscript a has been added to divide people into different age groups, as parameters like the mortality rate will likely vary substantially for people of different age groups. This way we can account for this to make our model more realistic. As a starting point we will divide people into 3 age groups of 0–17, 18–65, 65+ to limit the complicatedness of our model, but dividing people into smaller intervals of age is also possible at the cost of a higher computational load.

2.1 New parameters:

- μ_a : mortality rate for each age group a .
- ν_a : vaccination rate for each age group a .

2.2 New constants:

- $\omega_{r,a}$: chance of a recovered individual from age group a to become susceptible again in a time dt .
- $\omega_{v,a}$: chance of a vaccinated individual from age group a to become susceptible again in a time dt .
- C_{ab} : similar to the contact matrix W_{ij} , this is a matrix that contains information about the relative amount of contact between age groups. It should also be normalized: $\sum_b C_{ab} = 1$

2.3 Simulation:

We can now simulate our model in the same way as we have done before by introducing the chances of a person dying and a person getting vaccinated:

- $p_{death,a} = 1 - e^{-\mu_a \Delta t}$
- $p_{vac,a} = 1 - e^{-\nu_a \Delta t}$
- $p_{breakthrough,r,a} = 1 - e^{-\omega_{r,a} \Delta t}$
- $p_{breakthrough,v,a} = 1 - e^{-\omega_{v,a} \Delta t}$

In similar fashion to what we did before, we can then take binomial draws of:

- $(S_{i,a} - newinfections_{i,a}, p_{vac,a}) = newvaccinations_{i,a}$
- $(I_{i,a}, p_{death,a}) = newdeaths_{i,a}$
- $(R_{i,a}, p_{breakthrough,r,a}) = newbreakthrough_{r,i,a}$
- $(V_{i,a}, p_{breakthrough,v,a}) = newbreaktrhough_{v,i,a}$

$$\begin{cases} S_{i,a}(t + \Delta t) = S_{i,a}(t) - newinfections_{i,a} - newvaccinations_{i,a} + newbreakthroughs_{r,i,a} + newbreaktrhough_{v,i,a} \\ I_{i,a}(t + \Delta t) = I_{i,a}(t) + newinfections_{i,a} - newrecoveries_{i,a} - newdeaths_{i,a} \\ R_{i,a}(t + \Delta t) = R_{i,a}(t) + newrecoveries_{i,a} - newbreakthroughs_{r,i,a} \\ D_{i,a}(t + \Delta t) = D_{i,a}(t) + newdeaths_{i,a} \\ V_{i,a}(t + \Delta t) = V_{i,a}(t) + newvaccinations_{i,a} - newbreaktrhoughs_{v,i,a} \end{cases}$$

2.4 Limiting assumptions:

- We assume that only susceptible people get vaccinated and not people who have already been infected by the disease but don't show symptoms yet.

- As per the equations, the number $V_{i,a}$ now represents the amount of people immune to infection, not the total number of people vaccinated. This also means that after an individual has lost their immunity from vaccination, the model may select them to be vaccinated again. This is a shortcoming of our model that makes the waning dynamic of a vaccinated person possible, but when booster vaccinations are part of government policy, this actually does get closer to realism again.
- In our simulation, some people can die from the disease, but natural rates of birth and death are not taken into account in our model, which will have an unrealistic effect when simulating longer timescales (in the order of years). Aging of people is also not taken into account so there are no transition of people going from one age group to the next.
- People from the same age group and municipality mix homogeneously in our simulation.
- People become infectious to others immediately after having been infected themselves.
- The transmission rate β is the same for all age groups. The severity of symptoms an infected individual experiences also has no influence on the chance of them infecting another susceptible individual.